Test another explanation

Generate plausible alternatives and ask what could distinguish them.

Suggested teaching time: 30 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: None

Students will be able to

- Explain how more than one account could fit an observation.
- Use supplied details to propose plausible alternatives.
- Name evidence that could help distinguish two explanations.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; no student devices required.
- Instructor guide with model answers, kept separate from the handout.
- Read the three scenarios and their model answers.
- Print the three-page handout or display one activity at a time.
- Explain that all scenarios and figures are invented for reasoning practice.

Teaching sequence

5 min	**Offer a plausible alternative.** Use the cafe warm-up. Require a link between the new office and possible customers, not just a label.
10 min	**Explain three museum possibilities.** Pairs complete the museum activity, explaining how each supplied change could influence visits.
8 min	**Choose evidence that separates accounts.** Discuss model answers and prompts. Check that proposed comparisons vary the factor of interest rather than changing everything together.
7 min	**Test transfer independently.** Learners complete the library exit ticket. Assess whether their alternatives remain possibilities and their comparison is relevant.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / The cafe gets busier

Student task and instructor answer

Explain these words as needed

Alternative explanation: Another account that could fit the same observation.

Mechanism: A description of how a proposed cause could produce the outcome.

Scenario

Invented scenario: A cafe's lunchtime visits rise during the week it introduces music. A large office opens next door that same week. The owner credits the music.

Student output: Write one alternative explanation and one sentence stating what remains unknown.

1. Use the new office to give an alternative to the music explanation.

2. Explain how it could increase lunchtime visits.

3. State whether the facts show which explanation contributed more.

Student success check: Have you explained a connection while avoiding an unsupported verdict?

Model answer

The new office could bring nearby workers looking for lunch. The facts do not show how much the music or the office contributed, and both could matter.

Why this meets the task: The answer supplies a plausible mechanism and preserves uncertainty between compatible explanations.

Main activity / Why did museum visits rise?

Student task and instructor answer

Scenario

Invented scenario: A museum's family visits rise from 200 to 320 during the first weekend of free admission. That weekend also brings a new dinosaur exhibit, a school holiday, and unusually heavy rain. The museum credits free admission alone.

Student output: Write three possible explanations with reasons, then one proposed comparison for a chosen rival.

1. List three rival explanations using the supplied details.
2. For each, write one sentence explaining how it could increase museum visits. Mark these as possibilities, not established causes.
3. Choose one rival and name a comparison that would help distinguish it from the effect of price.

Student success check: Have you explained a plausible connection for each rival instead of merely listing coinciding events?

Model answer

The dinosaur exhibit might attract interested families; the holiday might give them more free time; rain might make indoor activities more attractive. To examine the exhibit explanation, compare similar weekends with the same exhibit but different admission prices while keeping holiday and weather conditions as comparable as possible. The supplied facts do not tell us which cause contributed how much.

Why this meets the task: It supplies mechanisms for three alternatives and a relevant follow-up comparison without presenting possibilities as findings.

Exit ticket / The library app launch

Student task and instructor answer

Scenario

Invented scenario: Library loans rise in the month a booking app launches. The library also extends opening hours and starts a popular reading challenge that month. The director credits the app alone.

Student output: Write two alternative explanations and one comparison that could help test the app.

1. Explain how longer opening hours could increase loans.

2. Explain how the reading challenge could increase loans.

3. Describe a comparison with and without the app that keeps the other two changes comparable.

Student success check: Are your explanations possibilities, and would the comparison separate the app from the other changes?

Model answer

Longer hours could let more people borrow; the challenge could motivate more borrowing. Compare similar branches over the same period, some with the app and some without, with comparable hours and the same challenge. These facts alone do not assign the increase to any one change.

Why this meets the task: The answer links each rival to borrowing and proposes a comparison that could provide more informative evidence.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- Alternatives use the supplied facts and fit the observed outcome.

- Each alternative includes a plausible connection rather than a mere coincidence.

- The proposed comparison could distinguish explanations without treating an untested possibility as a finding.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Must the rival explanations exclude one another?

Listen for: No. Free admission, an exhibit, holidays, and weather could all influence visits. Their separate contributions remain unknown.

Does imagining a possibility make it equally likely?

Listen for: No. A possibility needs a plausible connection and then evidence. Naming it does not establish its probability or importance.

Common confusions

Any story is a useful alternative. A useful alternative fits the known facts and explains how the outcome could occur.

Listing a coincidence explains its effect. State the mechanism: why might the event influence the outcome?

One plausible alternative settles the issue. It opens a question. Seek evidence that distinguishes accounts, and allow multiple factors to contribute.

Adapt the lesson

More support: Provide the frame: [change] might lead to [action], which could increase [outcome]. Use the museum facts to complete one example together. Expected example: the new office could supply more nearby lunchtime customers; this remains a possibility.

More challenge: Ask whether a rainy weekend with free admission compared with a dry paid weekend isolates price. Expected response: no; both price and weather differ, so either could help explain the difference.

Next practice

https://xixteen.com/learners/?practice=alternative_explanations#workbench